

Problem

Show that the transmission parameters of a two-port may be obtained from the y parameters as:

$$\mathbf{A} = -\frac{y_{22}}{y_{21}}, \quad \mathbf{B} = -\frac{1}{y_{21}}$$

$$\mathbf{C} = -\frac{\Delta y}{y_{21}}, \quad \mathbf{D} = -\frac{y_{11}}{y_{21}}$$

Step-by-step solution

Step 1 of 2

We know that Y parameter equations

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad -(1)$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad -(2)$$

$$\text{or,} \quad V_1 = \left(-\frac{Y_{22}}{Y_{21}} \right) V_2 + \left(-\frac{1}{Y_{21}} \right) (-I_2) \quad [\text{Using equation (2)}]$$

$$\equiv A V_2 + B(-I_2)$$

Step 2 of 2

Again

$$\begin{aligned} I_1 &= Y_{11}V_1 + Y_{12}V_2 \\ &= Y_{11} \left[\left(-\frac{Y_{22}}{Y_{21}} \right) V_2 + \left(-\frac{1}{Y_{21}} \right) (-I_2) \right] + Y_{12}V_2 \\ &= V_2 \left(\frac{Y_{12}Y_{23} - Y_{11}Y_{22}}{Y_{23}} \right) + \left(-\frac{Y_{11}}{Y_{21}} \right) (-I_2) \\ &= \left(\frac{-\Delta Y}{Y_{21}} \right) V_2 + \left(-\frac{Y_{11}}{Y_{21}} \right) (-I_2) \\ &\equiv C V_2 + D(-I_2) \end{aligned}$$

So, we have

$$A = -\frac{Y_{22}}{Y_{21}}; \quad B = -\frac{1}{Y_{21}} \quad C = -\frac{\Delta Y}{Y_{21}} \quad D = -\frac{Y_{11}}{Y_{21}}$$